

Asset Price Redistribution

Andreas Fagereng, Matthieu Gomez, Émilien Gouin-Bonenfant, Martin Holm, Benjamin Moll, Gisle Natvik

Presented by Tommaso De Santo

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New York University

Over the last decades, there have been increases in valuations across many asset classes.
Are there winners and losers in this process?

- One view: actual shift of resources to the wealthy (e.g Saez, Yagan and Zucman, 2021)
- Another opinion: these are just 'paper gains' (Cochrane 2020, Krugman 2021)

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What does the paper do?

- Theory:

- Develop a sufficient statistic approach that quantifies the individual welfare gain due to changes in asset prices
- Key features:
 - Transactions, not holdings, drive welfare
 - Asset prices are purely redistributive

- Empirics:

- Implement the formula above with very rich Norwegian data from 1994 to 2015
- Main finding: large redistributive effects
 - Young lose, old gain
 - Wealthy gain more than poor
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Theory

Baseline model

- Deterministic, infinite horizon
- B_t holdings of one period bonds with face value one and price Q_t
 - Holding a one period bond equivalent to investing in deposit with interest rate $R_{t+1} = 1/Q_t$
- $R_{0 \rightarrow t} = R_1 \cdot R_2 \cdot \dots \cdot R_t$ is the cumulative return
- $N_{k,t}$ holdings of long lived asset k that pays dividends $D_{k,t}$ and costs $P_{k,t}$
 - strictly convex adjustment costs $\chi_k(N_{k,t} - N_{k,t-1})$

Household problem and Welfare Gain

- Households have time separable preferences with a strictly concave $U(\cdot)$
- They receive income stream $\{Y_t\}_{t=0}^{\infty}$ and are price takers
- The household problem

$$V = \max_{\{C_t, B_t, \{N_{k,t}\}_k\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(C_t)$$

$$\text{s.t., } \forall t \quad C_t + \sum_{k=1}^K (N_{k,t} - N_{k,t-1})P_{k,t} + B_t Q_t + \sum_{k=1}^K \chi_k (N_{k,t} - N_{k,t-1}) = \sum_{k=1}^K N_{k,t-1} D_{k,t} + B_{t-1} + Y_t$$

Welfare Gain (cont.)

Proposition

Consider an arbitrary perturbation of the path of asset prices, $\{dQ_t, \{dP_{kt}\}_k\}_{t=0}^{\infty}$. The implied Welfare gain is

$$\text{Welfare Gain} \equiv \frac{dV}{U'(C_0)} = \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} \left(\sum_{k=1}^K (N_{k,t} - N_{k,t-1}) dP_{k,t} - B_t dQ_t \right)$$

Details

In dC_t terms

- Welfare gain depends on being a buyer or a seller, but timing matters
- For liquid assets, transactions=holdings, and $dQ_t > 0$ benefits households holding short term debt
- $\{\{D_{kt}\}_k\}_{t=0}^{\infty}$ is held constant and so price deviations are pure valuation effects
- We can think of it as a household's **willingness to pay** for the price deviation

Corollary

Suppose initial prices clear all asset markets. Then welfare gains implied by a price deviation aggregate to zero, so price deviations are purely redistributive.

- Obvious: if someone sells someone buys
- We are using a **money metric**: no social welfare function
- Can be extended to include other sectors

Implementation and sufficient statistic

For non infinitesimal deviations, our formula will be

$$\text{Welfare Gain} = \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} \left(\sum_{k=1}^K (N_{k,t} - N_{k,t-1}) \Delta P_{k,t} - B_t \Delta Q_t \right)$$

But what to do about dividends changing in the real world? Consider a baseline scenario where the price/dividend ratio is constant and construct price deviations from this baseline $\overline{PD}_k$:

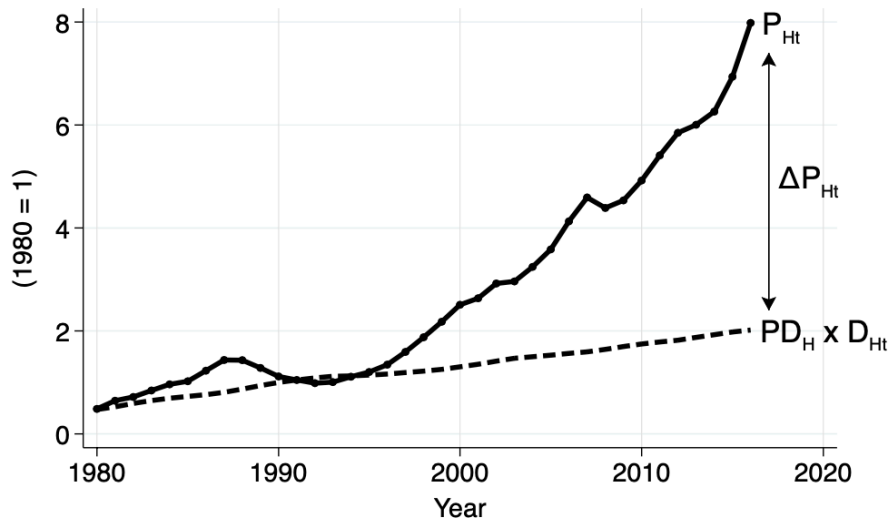
$$\Delta P_{k,t} = P_{k,t} - \overline{PD}_k \times D_{k,t}$$

or:

$$\frac{\Delta P_{k,t}}{P_{k,t}} = \frac{PD_{k,t} - \overline{PD}_k}{PD_{k,t}}$$

Discussion about discount shocks

Graphical representation of the price deviation ΔP_{Ht}



Combining the approximation with the empirical price deviations and truncating the formula we get

$$\text{Welfare Gain} = \sum_{t=0}^T R_{0 \rightarrow t}^{-1} \left(\sum_{k=1}^K (N_{k,t} - N_{k,t-1}) P_{k,t} \times \frac{PD_{k,t} - \overline{PD}_k}{PD_{k,t}} - B_t Q_t \times \frac{Q_t - \overline{Q}}{Q_t} \right)$$

For non infinitesimal deviations, it's an approximation, but the error is small Second order

Empirical implementation

Sample and asset classes

- 20 years from 1994 to 2015, individuals over 18
- Four main asset classes:

$$\text{Welfare Gain}_{\text{housing}} = \sum_{t=0}^{\infty} R^{-t} (N_{H,t-1} - N_{H,t}) P_{H,t} \times \frac{PD_{H,t} - \overline{PD}_H}{PD_{H,t}}$$

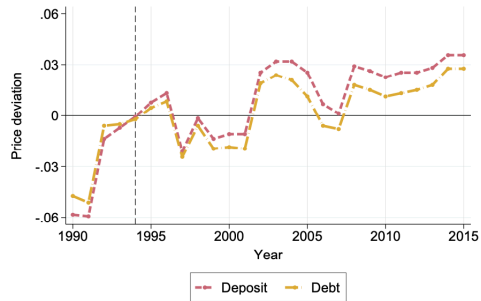
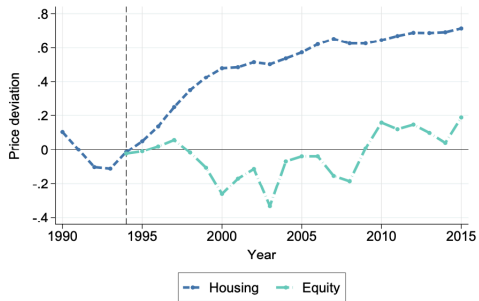
$$\text{Welfare Gain}_{\text{debt}} = \sum_{t=0}^{\infty} R^{-t} (-B_{M,t} Q_{M,t}) \times \frac{Q_{M,t} - \overline{Q}_M}{Q_{M,t}}$$

$$\text{Welfare Gain}_{\text{deposit}} = \sum_{t=0}^{\infty} R^{-t} (-B_{D,t} Q_{D,t}) \times \frac{Q_{D,t} - \overline{Q}_D}{Q_{D,t}}$$

$$\text{Welfare Gain}_{\text{equity}} = \sum_{t=0}^{\infty} R^{-t} (N_{E,t-1} - N_{E,t}) P_{E,t} \times \frac{PD_{E,t} - \overline{PD}_E}{PD_{E,t}}$$

- Interest rates: *Statistics Norway* database
- Price to rent-ratio: excellent private data but only from 2012. Combine with rental index from *Statistics Norway* and house price series from Eitrheim and Erlandsen (2004)
- Equity valuation: aggregate measure of cash flows over enterprise value from *Worldscope*.

Price deviations



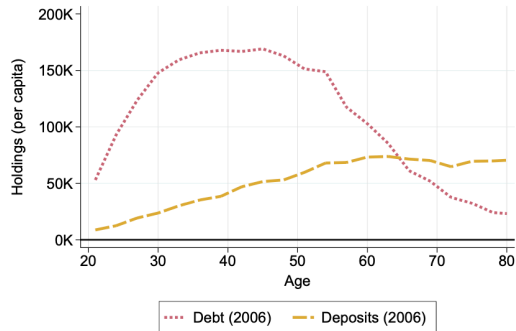
Combination of Norwegian administrative registries: uncensored individual data for tax payments, equity shares, balance sheets, housing transactions.

- **Holdings:** very detailed data on many forms of holdings
 - For housing, combine data and registered housing characteristics
 - For households owning a private business, computation of indirect shares
- **Transactions:**
 - Housing transactions are directly observed
 - Public equities: impute transactions based on beginning and end of the year holdings
 - Private businesses: assumed illiquid. But a lot of attention to exposure to change in interest rates

Financial holdings and transactions by age group (2006)



(a) Housing and equity



(b) Debt and deposits

Empirical findings

Intuitively our welfare gain measure is

$$\text{Welfare Gain}_i = \sum_{t=0}^T R^{-t} (\text{Sales}_{it} \times \text{Price Deviation}_t)$$

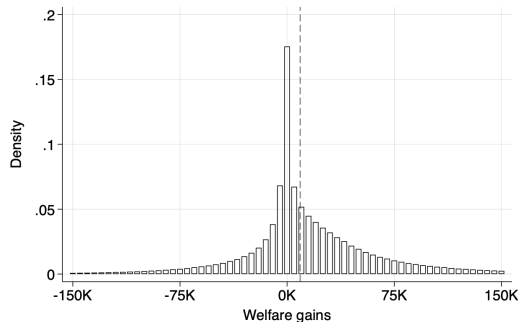
- We apply it and find large redistributive effects
 - young lose, old gain
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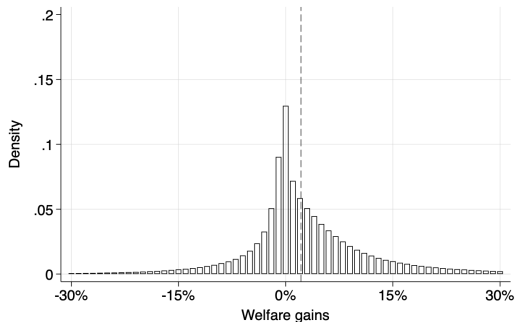
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Distribution of welfare gains



(a) Welfare gains in dollar terms

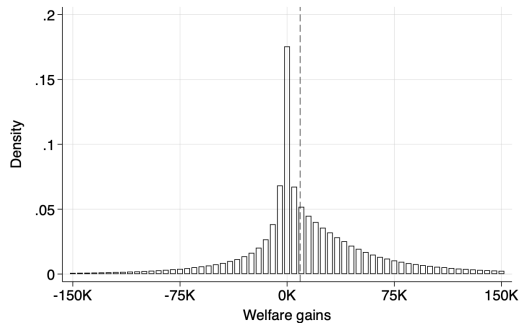


(b) Welfare gains as percent of total wealth

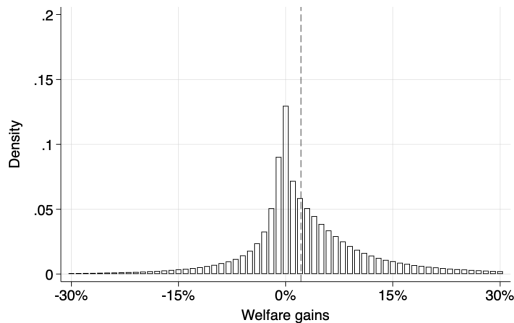
- -35k at 10p, 80k at 90p
- -280k at 1p, 1m at 99p

- -6% at 10p, +15 % at 90p
- -67% at 1p, +152 % at 99p

Distribution of welfare gains



(a) Welfare gains in dollar terms

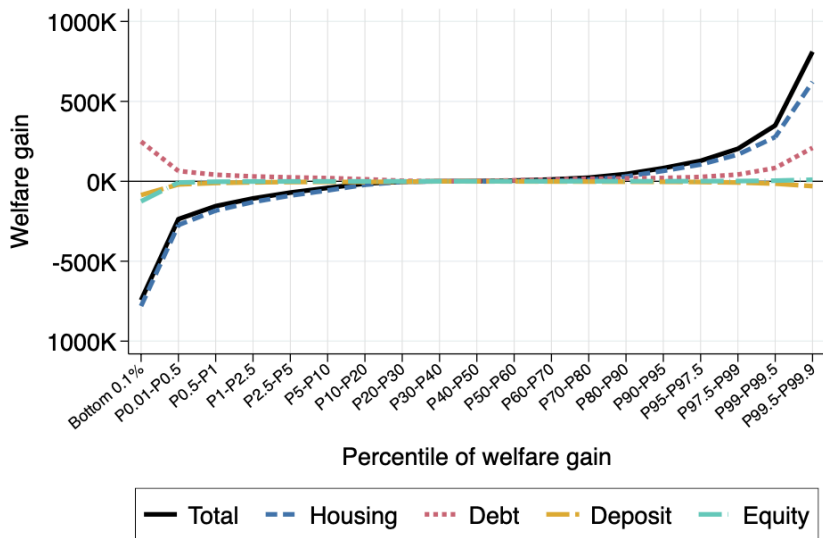


(b) Welfare gains as percent of total wealth

- -35k at 10p, 80k at 90p
- -280k at 1p, 1m at 99p

- Kelly skewness 0.33, kurtosis 9

Decomposition of welfare gains by percentile

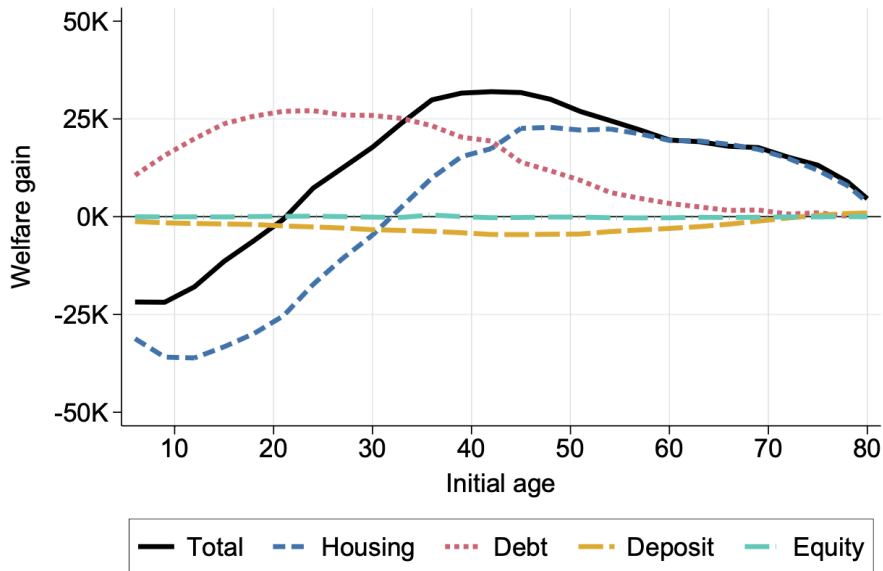


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Welfare across cohorts



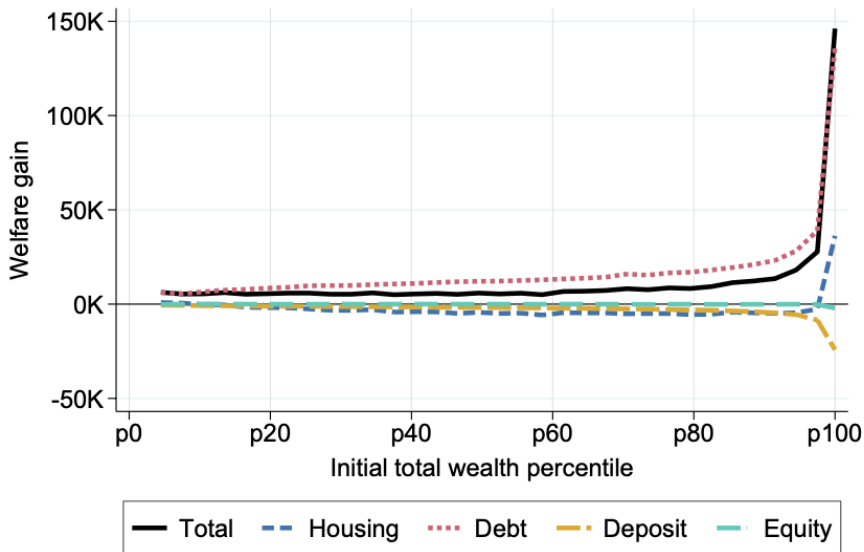
Main findings

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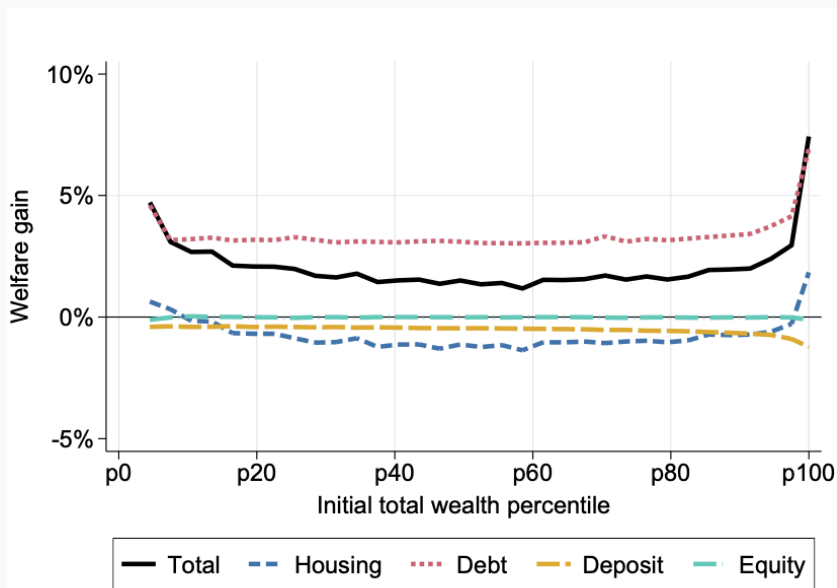
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Welfare gains across wealth percentiles



Welfare gains across wealth percentiles: normalized by initial wealth



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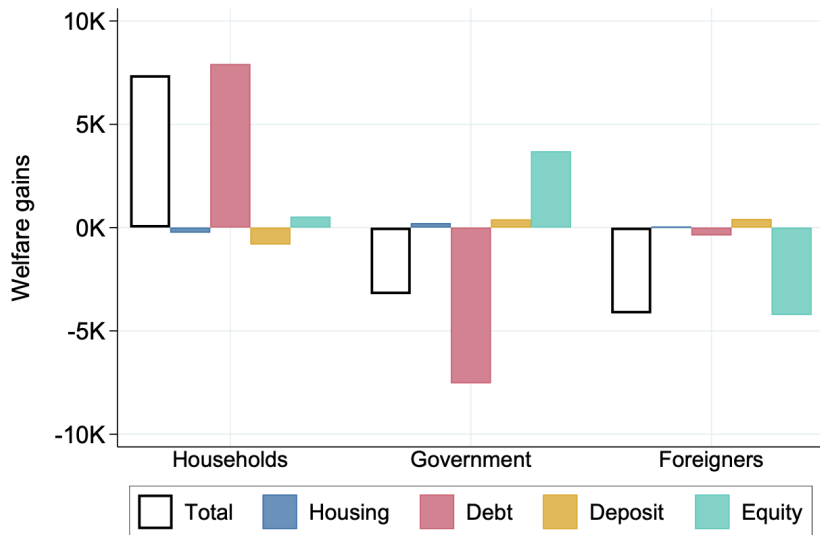
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- When we add non-households entities, we have

$$\text{Welfare Gain}_H + \text{Welfare Gain}_G + \text{Welfare Gain}_F = 0$$

- Use data from Financial accounts:
 - Consolidate government with non-profit (big role played by a sovereign wealth fund)
 - Integrate Financial accounts with housing registry data
- Main flow trends:
 - Government is a net buyer of foreign equities
 - Household has a large and growing level of debt, ultimately held by the government: a decline in interest rates will redistribute from government towards households

Welfare gains across sectors



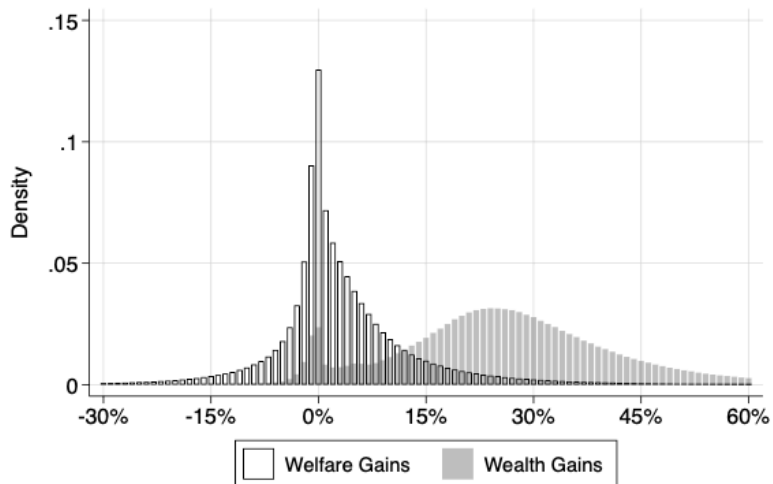
Welfare versus wealth gains

- Wealth gains are deviations in wealth due to deviations in asset prices:

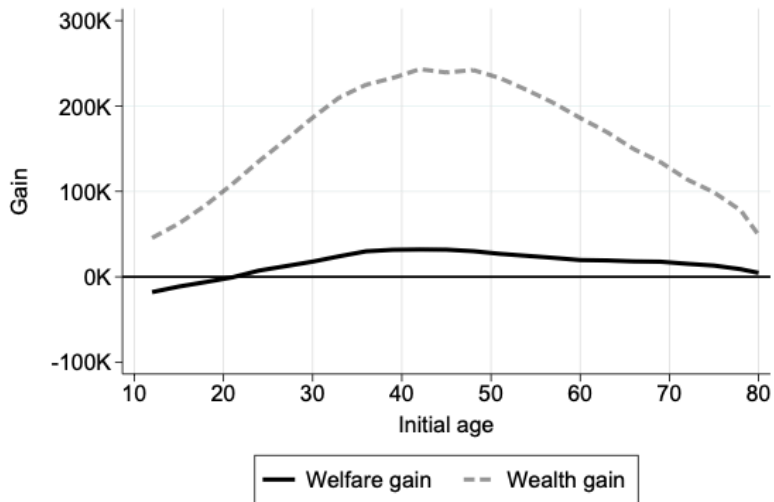
$$\text{Wealth gain} = \sum_{t=0}^T R_{0 \rightarrow t}^{-1} \sum_{k=1}^K N_{k,t-1} (\Delta P_{k,t} - R_t \Delta P_{k,t-1}) - \sum_{t=1}^T R_{0 \rightarrow t}^{-1} B_t \Delta Q_t$$

- For the one period bond, welfare and wealth contribution will be the same
- For long-lived assets, it's different: wealth gains don't capture negative future returns for $t > T$ [Details](#)

Welfare versus wealth gains: households



Welfare versus wealth gains: cohorts



Two robustness checks provided:

- **Second order approximation:** effects seem to be quantitatively similar, biggest change is lower gain from debt [Details](#)
- **Extrapolated valuation changes**
 - Truncating our formula means assuming that prices revert to the baseline after 2015
 - Counterfactual: assume valuations remain at 2015 level and future generations behave the same adjusting for growth
 - Main result is that the unborn are negatively affected [Details](#)

Conclusions

- The paper develops a simple framework to quantify welfare effects of asset price fluctuations
- Big redistributive effects from Norway data:
 - From young to old
 - From poor to wealthy
 - From government to households
- Possible implications for policy?
 - Taxing wealth or capital gains does not seem optimal
 - Will the young suffer more in the future due to the government's negative gain?

Appendix

Extensions

Many different extensions:

- Stochastic environment
- Borrowing and collateral
- Bequests
- General equilibrium
- Government sector
- Housing and wealth in the utility function
- Relation to duration mismatch

Intuition: a two period model

- Two periods, $t = 0, 1$, income Y_0, Y_1
- Asset available for trade, amount N . Households maximize

$$V = \max_{C_0, C_1, N_0} U(C_0) + \beta U(C_1)$$

subject to

$$C_0 + (N_0 - N_{-1})P_0 = Y_0$$

$$C_1 = N_0 D_1 + Y_1$$

- With a small variation dP_0 , holding D_1 constant, we have

$$dV = U'(C_0)(N_{-1} - N_0)dP_0$$

Welfare versus wealth gains

The Welfare gain measure:

$$\frac{dV}{U'(C_0)} = (N_{-1} - N_0)dP_0$$

How can asset holders not benefit from rise in prices if their wealth increases?

- $R_0 = \frac{P_0}{P_{-1}}, R_1 = \frac{D_1}{P_0}.$
- $\frac{dR_0}{dP_0} = \frac{1}{P_{-1}} > 0, \frac{dR_1}{dP_0} = -\frac{R_1}{P_0} < 0$
- So $P_0 \uparrow \implies R_0 \uparrow$ but $R_1 \downarrow$, and WG depends on net sales

Proof of proposition 1

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t U(C_t) + \sum_{t=0}^{\infty} \lambda_t \left(\sum_{k=1}^K N_{k,t-1} D_t + B_{t-1} + Y_t - C_t - \sum_{k=1}^K (N_{k,t} - N_{k,t-1}) P_{k,t} - \sum_{k=1}^K \chi_k (N_{k,t} - N_{k,t-1}) - B_t Q_t \right)$$

The first order conditions are

$$\begin{aligned} \beta^t U'(C_t) &= \lambda_t & (\partial \mathcal{L} / \partial C_t &= 0) \\ \lambda_t (\chi'_k (N_t - N_{t-1}) + P_t) &= \lambda_{t+1} (D_{t+1} + \chi'_k (N_{t+1} - N_t) + P_{t+1}) & (\partial \mathcal{L} / \partial N_{kt} &= 0) \\ \lambda_t Q_t &= \lambda_{t+1} & (\partial \mathcal{L} / \partial B_t &= 0) \end{aligned}$$

Using the Envelope theorem, the infinitesimal change in the value function is given by the infinitesimal change in the Lagrangian:

$$\begin{aligned}
 dV &= \sum_{t=0}^{\infty} \left(\sum_{k=1}^K \frac{\partial \mathcal{L}}{\partial P_{k,t}} dP_{k,t} + \frac{\partial \mathcal{L}}{\partial Q_t} dQ_t \right) \\
 &= \sum_{t=0}^{\infty} \lambda_t \left(\sum_{k=1}^K (N_{k,t-1} - N_{k,t}) dP_{k,t} - B_t dQ_t \right) \\
 &= \lambda_0 \sum_{t=0}^{\infty} (Q_0 \dots Q_{t-1}) \left(\sum_{k=1}^K (N_{k,t-1} - N_{k,t}) dP_{k,t} - B_t dQ_t \right) \\
 &= U'(C_0) \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} \left(\sum_{k=1}^K (N_{k,t-1} - N_{k,t}) dP_{k,t} - B_t dQ_t \right)
 \end{aligned}$$

Welfare gain in terms of consumption

- Consider an arbitrary perturbation of asset prices, $\{dQ_t, \{dP_{kt}\}_k\}_{t=0}^{\infty}$
- with dV the price deviation effect, define

$$\text{Welfare Gain} \equiv \frac{dV}{U'(C_0)}$$

- Totally differentiating V we can write

$$\text{Welfare Gain} = \sum_{t=0}^{\infty} \beta^t \frac{U'(C_t)}{U'(C_0)} dC_t = \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} dC_t$$

- We can think of it as household's **willingness to pay** for the price deviation

Interpreting the price deviations as discount shocks

- $\mathcal{R}_t$ is the discount rate from t to $t + 1$, $\{D_t, \mathcal{R}_t\}$ given
- Suppose the sequence of asset prices $\{P_t\}_{t=0}^{\infty}$ satisfies $P_t = \frac{D_{t+1} + P_{t+1}}{\mathcal{R}_t}$

- So

$$P_t = \frac{D_{t+1}}{\mathcal{R}_t} + \frac{D_{t+2}}{\mathcal{R}_t \mathcal{R}_{t+1}} + \dots = \sum_{s=1}^{\infty} \mathcal{R}_{t \rightarrow t+s}^{-1} D_{t+s}$$

- Strong assumption: dividends grow at a constant rate G , $D_{t+s} = D_t G^s$. Then

$$P_t = D_t \sum_{s=1}^{\infty} \mathcal{R}_{t \rightarrow t+s}^{-1} G^s$$

- Or $PD = \sum_{s=1}^{\infty} \mathcal{R}_{t \rightarrow t+s}^{-1} G^s$

Interpreting the price deviations as discount shocks (cont.)

- If discount rates are constant, $\mathcal{R}_t = \mathcal{R}$ for all t , $\mathcal{R} > G$, then

$$\bar{P}_t = D_t \sum_{s=1}^{\infty} \mathcal{R}^{-s} G^s \quad \text{or} \quad \bar{P}_t = D_t \times \bar{PD} \quad \text{with} \quad \bar{PD} = \frac{G}{\mathcal{R} - G}$$

- In our empirical implementation, we construct price deviations as $\Delta P_t = P_t - \bar{PD} \times D_t$, or

$$\frac{\Delta P_t}{P_t} = \frac{PD_t - \bar{PD}}{PD_t}$$

- Combine them to get

$$\frac{\Delta P_t}{P_t} = \frac{\sum_{s=1}^{\infty} (\mathcal{R}_{t \rightarrow t+s}^{-1} - \mathcal{R}^{-s}) G^s}{\sum_{s=1}^{\infty} \mathcal{R}_{t \rightarrow t+s}^{-1} G^s}$$

Second order approximation

- Formally model a non-infinitesimal deviation in asset prices $\theta \in [0, 1]$, where $\{P_t(0)\}_{t=0}^{\infty}$ corresponds to the initial price path and $\{P_f(1)\}_{t=0}^{\infty}$ corresponds to the perturbed price path.
- Rewrite Welfare Gain =
$$\int_0^1 \sum_{t=0}^{\infty} R_{0 \rightarrow t}(\theta)^{-1} \left(\sum_{k=1}^K (N_{k,t-1}(\theta) - N_{k,t}(\theta)) dP_{k,t}(\theta) - B_t(\theta) dQ_t(\theta) \right) d\theta$$

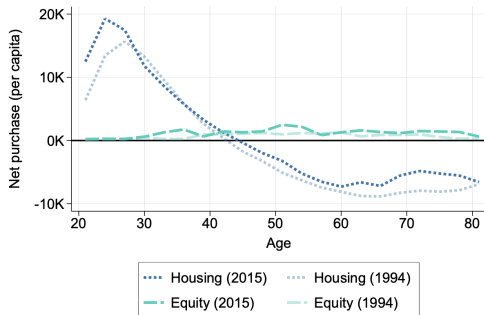
A trapezoidal approximation gives the second-order approximation

$$\begin{aligned} \text{Welfare Gain} &\approx \sum_{t=0}^{\infty} \left(R_{0 \rightarrow t} + \frac{\Delta R_{0 \rightarrow t}}{2} \right)^{-1} \\ &\quad \times \left(\sum_{k=1}^K \left(N_{k,t-1} - N_{k,t} + \frac{\Delta (N_{k,t-1} - N_{k,t})}{2} \right) \Delta P_{k,t} - \left(B_t + \frac{\Delta B_t}{2} \right) \Delta Q_t \right) \end{aligned}$$

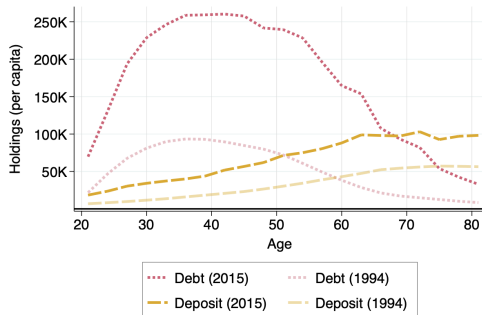
- This second-order approximation takes into account the welfare gains and losses due to asset transactions responding to asset price changes.
- To implement it, we need to assume how financial transactions would have changed if valuations didn't.
- Assumption: had valuations remained at 1994 level, agents of age n in 2015 would have behaved as agents of age n in 1994 after accounting for growth. Formally:

$$N_{k,t-1} - N_{k,t} + \Delta(N_{k,t-1} - N_{k,t}) = G^t(N_{k,-1} - N_{k,0})$$
$$B_t + \Delta B_t = G^t B_0$$

Transactions in 1994 versus 2015, adjusted for growth



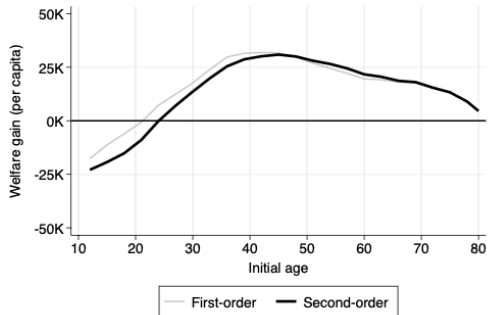
(a) Housing and equity



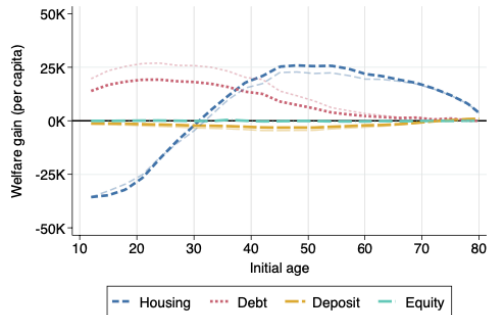
(b) Debt and deposits

- (a) plots $(N_{k,T} - N_{k,T-1})P_{k,T}$ and $G^T(N_{k,0} - N_{k,-1})P_{k,T}$. Remarkably close!
- (b) plots $B_T Q_T$ and $G^T B_0 Q_T$. Big difference is debt

Second-order approximation of welfare gains



(a) Total



(b) Detail by asset class

Extensions: stochastic environment

- Two asset model
- $p_t = \bar{p} + u_t$, $q_t = \bar{q} + v_t$, $d_t = \bar{d} + w_t$, where $\{u_t, v_t, w_t\}_{t=0}^{\infty}$ is a sequence of stochastic shocks.
- The welfare gain is given by

$$\text{Welfare Gain} \approx \sum_{t=0}^{\infty} ((N_{t-1} - N_t)P_t u_t - B_t Q_t v_t)$$

Extensions: borrowing and collateral constraints

- Two asset model with borrowing constraint $B_t \geq -F_t(N_t, P_t)$.
- The welfare gain is

$$\text{Welfare Gain} = \sum_{t=0}^{\infty} \frac{\beta^t U'(C_t)}{U'(C_0)} ((N_{t-1} - N_t) dP_t - B_t dQ_t) + \sum_{t=0}^{\infty} \frac{\mu_t}{U'(C_0)} F_p(N_t, P_t) dP_t$$

where $\mu_t \geq 0$ is the shadow value of relaxing the borrowing constraint.

- Asset appreciations could be beneficial for everybody if they move enough people off their borrowing constraints

Extensions: bequests

- Two asset model, addition of bequest I_t with warm-glow $F(.)$
- The welfare gain is

$$\text{Welfare Gain} = U'(C_0) \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} ((N_{t-1} - N_t + I_t) dP_t - B_t dQ_t) + U'(C_0) \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} P_t dI_t$$

- Two changes:
 - The net inheritance term $N_{t-1} - N_t + I_t$
 - The additional term incorporates readjustments

Extensions: general equilibrium

- A simple OLG with a tree, two cohorts
- The welfare gains for a demand shock $d\beta_0$

$$\text{Welfare Gain}_{-1} = (N_{-1})dP_0$$

$$\text{Welfare Gain}_0 = N_0dP_0 + \lambda_0^{-1} \log(C'_0/C_0)d\beta_0$$

Extensions: government sector

- Two asset model, government makes targeted transfer to households.
- The government acts to equalize the returns on assets,

$$Q_t^{-1} = \frac{D_{t+1} + P_{t+1} - \chi' (N_{t+1} - N_t)}{P_t + \chi' (N_t - N_{t-1})}$$

- The welfare gain is

$$\text{Welfare Gain}_i = \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} ((N_{it-1} - N_{it}) dP_t - B_{it} dQ_t) + \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} dT_{it}.$$

Extensions: housing and wealth in the utility function

- Have $U(C_t, F(N_t, P_t))$
- The welfare gain is

$$\text{Welfare Gain} = \sum_{t=0}^{\infty} R_{0 \rightarrow t}^{-1} ((N_{t-1} - N_t) dP_t - B_t dQ_t) + \sum_{t=0}^{\infty} \beta^t \frac{U_F(C_t, F(N_t, P_t))}{U_C(C_0, F(N_0, P_0))} F_P(N_t, P_t) dP_t$$

- Additional term accounts for direct effect of price deviations on utility

Extensions: relation to duration mismatch

- Households can trade bonds of all maturities
- Problem is

$$V = \max_{\{C_t, N_t, B_t\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t U(C_t)$$

subject to

$$C_0 + \sum_{h=1}^{\infty} B_h Q_{h0} = N_{-1} D_0 + Y_0$$

$$C_t = N_{-1} D_t + B_h + Y_t$$

Extensions: relation to duration mismatch (cont.)

$$\begin{aligned}\text{Welfare Gain} &= - \sum_{h=1}^{\infty} B_h \, dQ_{h0} \\ &= \sum_{h=1}^{\infty} (N_{-1}D_h + Y_h - C_h) \, dQ_{h0} \\ &= \sum_{h=1}^{\infty} R_{0 \rightarrow h}^{-1} (C_h - Y_h - N_{-1}D_h) \, d \log R_{0 \rightarrow h}\end{aligned}$$

- This corresponds to Appendix formula (A.37) in Auclert (2019)
- In the special case in which the perturbation is a level shift in the yield curve (i.e., $d \log R_0 = h d \log R$ for $h > 1$), the formula simplifies to

$$\text{Welfare Gain} = \left(\sum_{h=1}^{\infty} R_{0 \rightarrow h}^{-1} (C_h - Y_h - N_{-1}D_h) h \right) d \log R$$

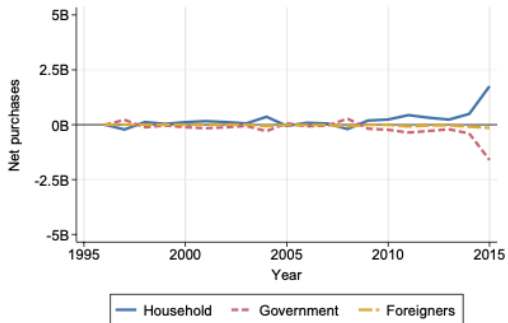
Household wealth in 1993

Table 1: Household wealth in 1993 (thousands of USD)

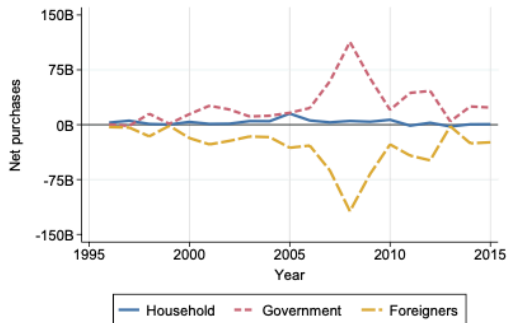
	Mean	S.D.	P10	P50	P90	P99
Total wealth	556.12	725.57	190.72	522.98	894.13	1,633.28
Financial and real wealth	125.63	649.84	-9.76	88.75	276.37	765.50
Housing	130.57	218.21	0.00	109.78	272.24	625.35
Debt	43.79	131.81	0.00	20.54	110.81	264.52
Deposits	22.62	93.97	0.05	6.66	56.39	211.83
Public equity	2.70	431.41	0.00	0.00	0.18	24.42
Private equity	9.17	419.39	0.00	0.00	0.00	121.35
Human wealth	430.49	280.82	98.91	429.01	722.89	1,220.19
Present value of labor income	323.29	310.89	0.00	314.90	673.86	1,169.33
Present value of net transfers	107.20	98.84	13.20	75.59	245.09	411.20

Notes. The total number of observations is 3,270,273. Values are reported in thousands of 2011 US dollars. Each statistic is computed for each variable separately.

Financial transaction by sector

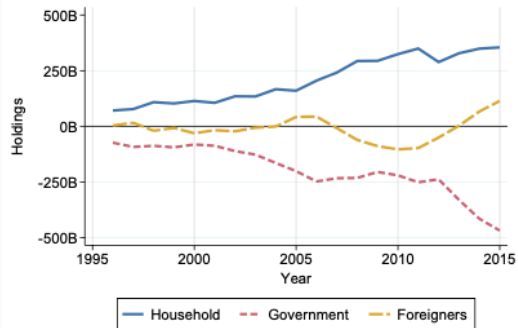


(a) Housing

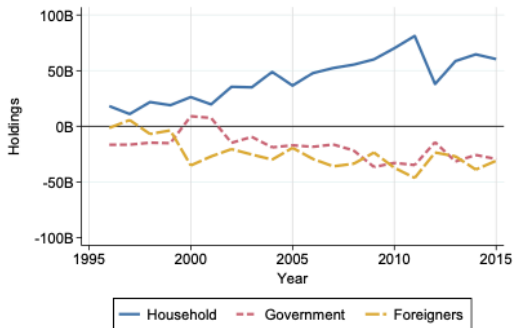


(b) Equity

Financial transaction by sector (cont.)



(c) Debt



(d) Deposits

Welfare versus wealth gains

Consider a variation in prices $(dP_{k,t})_{0 \leq t \leq T}$. For an asset $1 \leq k \leq K$, welfare gain equals wealth gains up to T adjusted for the effect of valuations on the terminal return at $T+1$:⁴¹

$$\underbrace{\sum_{t=0}^T R_{0 \rightarrow t}^{-1} (N_{k,t-1} - N_{k,t}) dP_{k,t}}_{\text{Welfare gain}} = \underbrace{\sum_{t=0}^T R_{0 \rightarrow t}^{-1} N_{k,t-1} P_{k,t-1} dR_{k,t}}_{\text{Wealth gain}} + \underbrace{R_{0 \rightarrow T+1}^{-1} N_{k,T} P_{k,T} dR_{k,T+1}}_{\text{Effect of } T+1 \text{ return}}.$$

where $\{dR_{k,t}\}_{t \geq 0}$ denotes the perturbation in asset returns corresponding to the perturbation in prices $\{dP_t\}_{t \geq 0}$; that is,

$$dR_{k,t} = (dP_{k,t} - R_t dP_{k,t-1}) / P_{k,t-1}.$$

To prove it, using summation by part, welfare gains for asset k can be rewritten as:

$$\begin{aligned}\sum_{t=0}^T R_{0 \rightarrow t}^{-1} (N_{k,t-1} - N_{k,t}) dP_{k,t} &= \sum_{t=0}^T R_{0 \rightarrow t}^{-1} N_{k,t-1} (dP_{k,t} - R_t dP_{k,t-1}) - R_{0 \rightarrow T}^{-1} N_{k,T} dP_{k,T} \\ &= \sum_{t=0}^T R_{0 \rightarrow t}^{-1} N_{k,t-1} P_{k,t-1} dR_{k,t} + R_{0 \rightarrow T+1}^{-1} N_{k,T} P_{k,T} dR_{k,T+1}\end{aligned}$$

To empirically implement

$$\begin{aligned}\text{Wealth Gain} &= \sum_{t=0}^T R_{0 \rightarrow t}^{-1} \sum_{k=1}^K N_{k,t-1} P_{k,t-1} \times \left(\frac{P_{k,t}}{P_{k,t-1}} \frac{PD_{k,t} - \overline{PD}_k}{PD_{k,t}} - R_t \frac{PD_{k,t-1} - \overline{PD}_k}{PD_{k,t-1}} \right) \\ &\quad - \sum_{t=0}^T R_{0 \rightarrow t}^{-1} B_t Q_t \times \frac{\Delta Q_t}{Q_t}.\end{aligned}$$

Estimate $P_{k,t}/P_{k,t-1}$ as $PD_{k,t}/PD_{k,t-1} \times G$, where $G = 1\%$, the growth rate of the Norwegian economy per capita, approximates the growth rate of dividends. [Back](#)

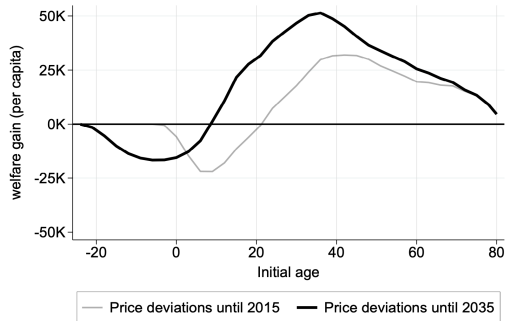
Extrapolated valuation changes

- Our main formula gives gain from with a path of price deviations that ends in 2015
- Alternative: assume valuations remaining permanently at their 2015 level.
- Need to impute the transactions in future years. Assume future transactions in each cohort will equal the transaction of the cohort with the same age in 2015, after adjusting for economic growth. 1994vs2015
- Formally, we assume that, for $t \geq T$, we have:

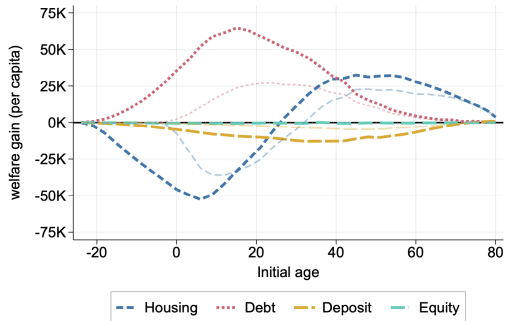
$$\begin{aligned}(N_{k,t} - N_{k,t-1}) P_{k,t} &= G^{t-T} (N_{k,T} - N_{k,T-1}) P_{k,T}, \\ B_t Q_t &= G^{t-T} B_T Q_T,\end{aligned}$$

where $(N_{k,T} - N_{k,T-1}) P_{k,T}$ and $B_T Q_T$ denote asset transactions and bond holdings in year $T = 2015$ and $G = 1.01$ denotes the real per-capita growth rate of the economy in our sample period.

Extrapolated valuation changes: results



(a) Total



(b) Detail by asset classes